KLH UNIVERSITY · GROUP PRESENTATION

Monthly Expenses Budget Analyzer

A beginner-friendly Java project for understanding money, logic, and modular programming

3–4 students · CO1 · CO2 · CO3



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One real problem, three learning outcomes

THE PROBLEM

Spending is scattered.

Monthly spending is scattered across small decisions and hard to compare with a budget.

Goal: turn income, budget, categories, and expenses into one readable report.

CO1

Frames the problem and algorithm.

CO2

Adds decisions, repetition, and safe exits.

CO3

Organizes the solution with methods, arrays, search, and summaries.

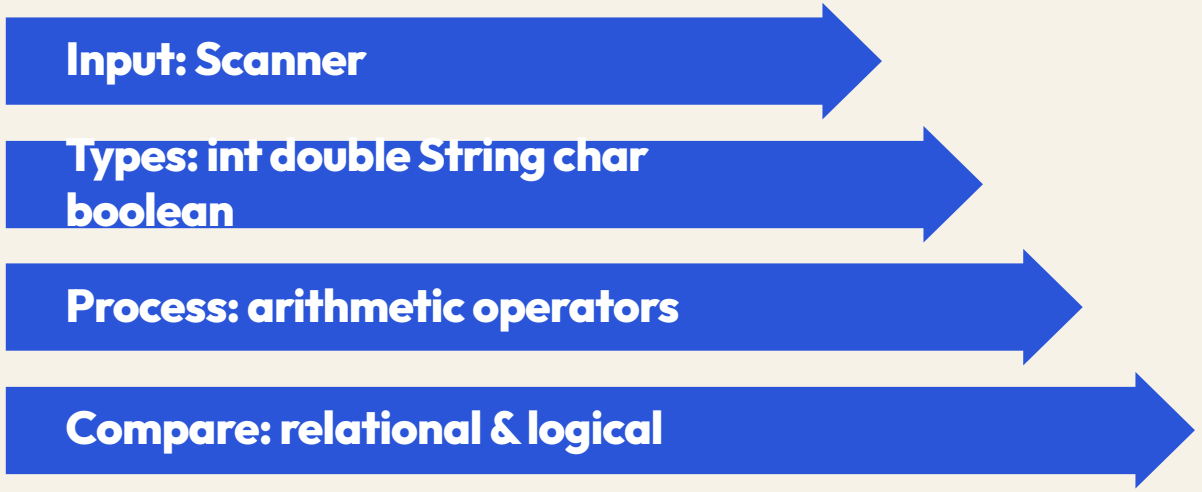
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CO1: from everyday problem to Java plan

01	02	03	04
Inputs	Process	Outputs	Algorithm
income, budget, number of expenses, category, description, amount	store values, calculate totals, compare percentages, classify status	expense list, remaining budget, average, highest expense, final report	read → validate → store → calculate → compare → display

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CO1 also makes the data visible



int	double	String	char	boolean
counts	money	labels	menu choices	checks

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CO2: the program decides, repeats, and protects the input



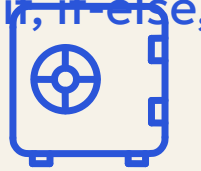
Decide

if, if-else, if-else-if



Repeat

for, while, do-while



Protect

continue and validation



Exit

break and switch

SPENDING STATUS

< 70%	70–100%	> 100%
Under control	Approaching limit	Exceeded

switch selects categories; for processes expenses; while/do-while repeats the menu.

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CO3: methods and arrays make the project explainable

Collect

inputIncome()
setBudget()
addExpense()

Summarize

calculateTotal()
calculateAverage()
findHighestExpense()

Answer

searchExpense()
countCategoryExpenses()
displayReport()

double[]

expenses

String[]

categories

String[]

descriptions

> Methods give each question a name. Arrays keep matching values organized.

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The Java code shows all three COs together

```
Scanner input = new Scanner(System.in);
double total = calculateTotal(expenses, count);
double remaining = budget - total;
double percentage = (total / budget) * 100;

if (percentage < 70) status = "Budget under
control";
else if (percentage <= 100) status =
"Approaching budget limit";
else status = "Budget exceeded";
```

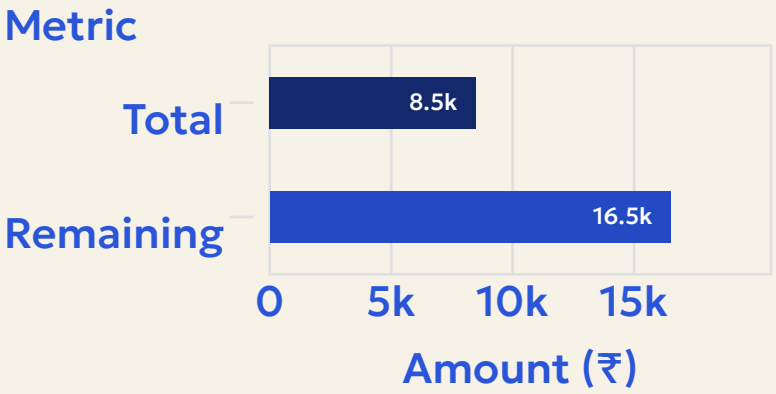
1. **CO1**
   Scanner, double, arithmetic, comparisons.
2. **CO2**
   if / else-if decision logic.
3. **CO3**
   Reusable calculateTotal() method and arrays.

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Demo: entries become a monthly financial report

Category	Amount
Food	₹500
Transportation	₹1,000
Education	₹3,000
Shopping	₹2,500
Entertainment	₹1,500

INCOME ₹30,000 · BUDGET ₹25,000



Total: ₹8,500 · **Remaining:** ₹16,500
Average: ₹1,700 · **Highest:** ₹3,000

Status: Budget under control

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Testing proves the logic at the edges

Test	Condition	Expected result
1	Expenses below budget	Under control
2	Total between 70% and 100%	Approaching limit
3	Total above budget	Budget exceeded
4	Negative or invalid amount	Skip and continue
5	Search by category or description	Match displayed

Advantages

simple console flow · reusable methods · clear monthly summary

Limits

manual input · in-memory arrays · one-month view

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Conclusion: a complete programming cycle in one small project

01

CO1

Problem → inputs → algorithm → flowchart

02

CO2

Decisions → repetition → validation → menu control

03

CO3

Methods → arrays → testing → explanation

04

Report

One readable monthly financial summary

GROUP SPLIT

problem + algorithm · CO2 logic · CO3 structure · demo + testing

FUTURE SCOPE

file storage · charts · multiple months · login · mobile or web interface

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